

Project Overview

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Jenseiroku (meaning a "Record of Life" in Japanese) is a privacy focused system that provides client side document encryption, subscription tracking, task management, life timeline, and smart reminder into a single unified platform.

At its core, Jenseiroku is built around the principle that personal data should never leave the user's device unprotected. Every document, note, and record stored within the platform is encrypted entirely on the client side using industry-standard cryptographic algorithms before any data is persisted or synced — meaning even the platform itself has zero knowledge of the user's content. This architecture ensures that sensitive financial records, personal timelines, and private documents remain completely inaccessible to third parties, advertisers, or potential breaches, giving users full ownership and sovereignty over their own life data.

Architecture Overview

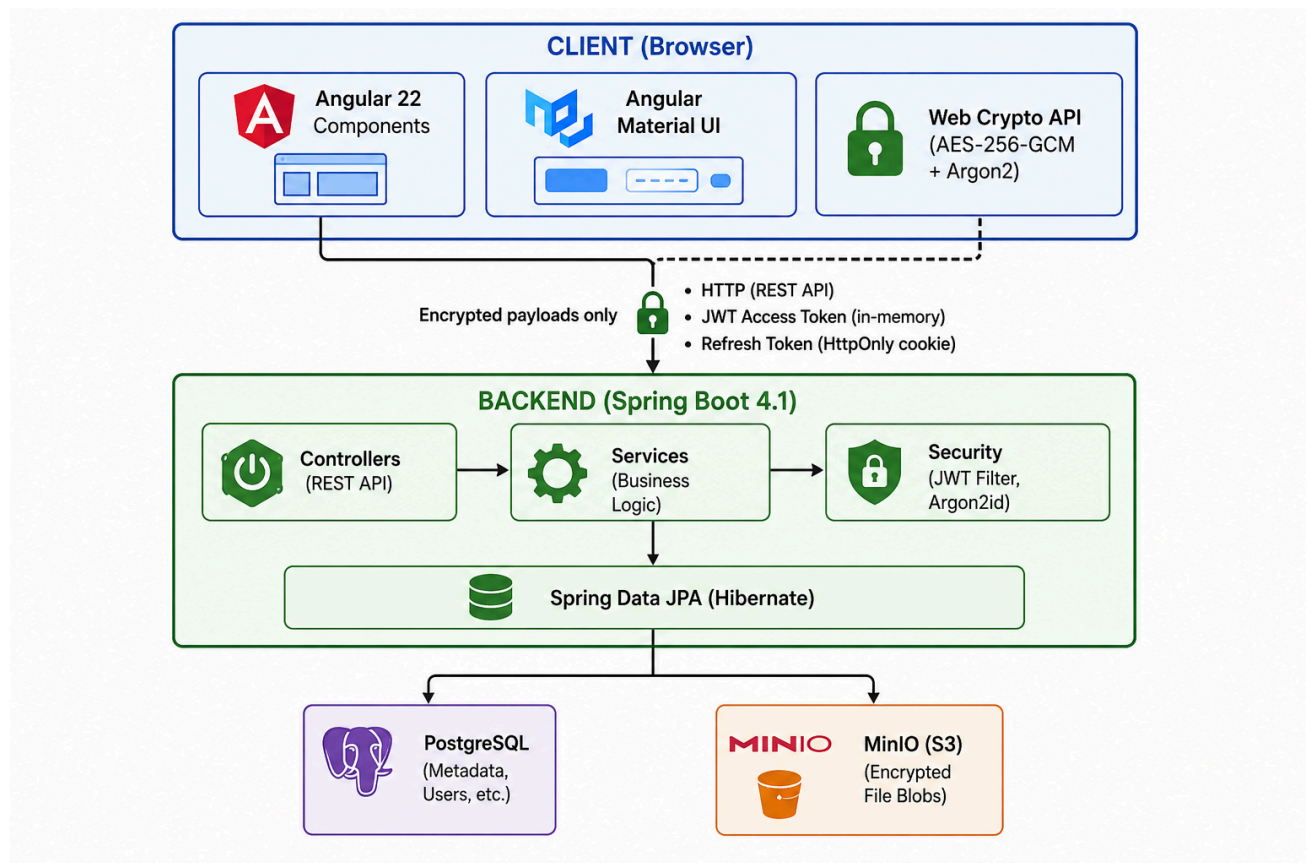


Fig: architecture overview

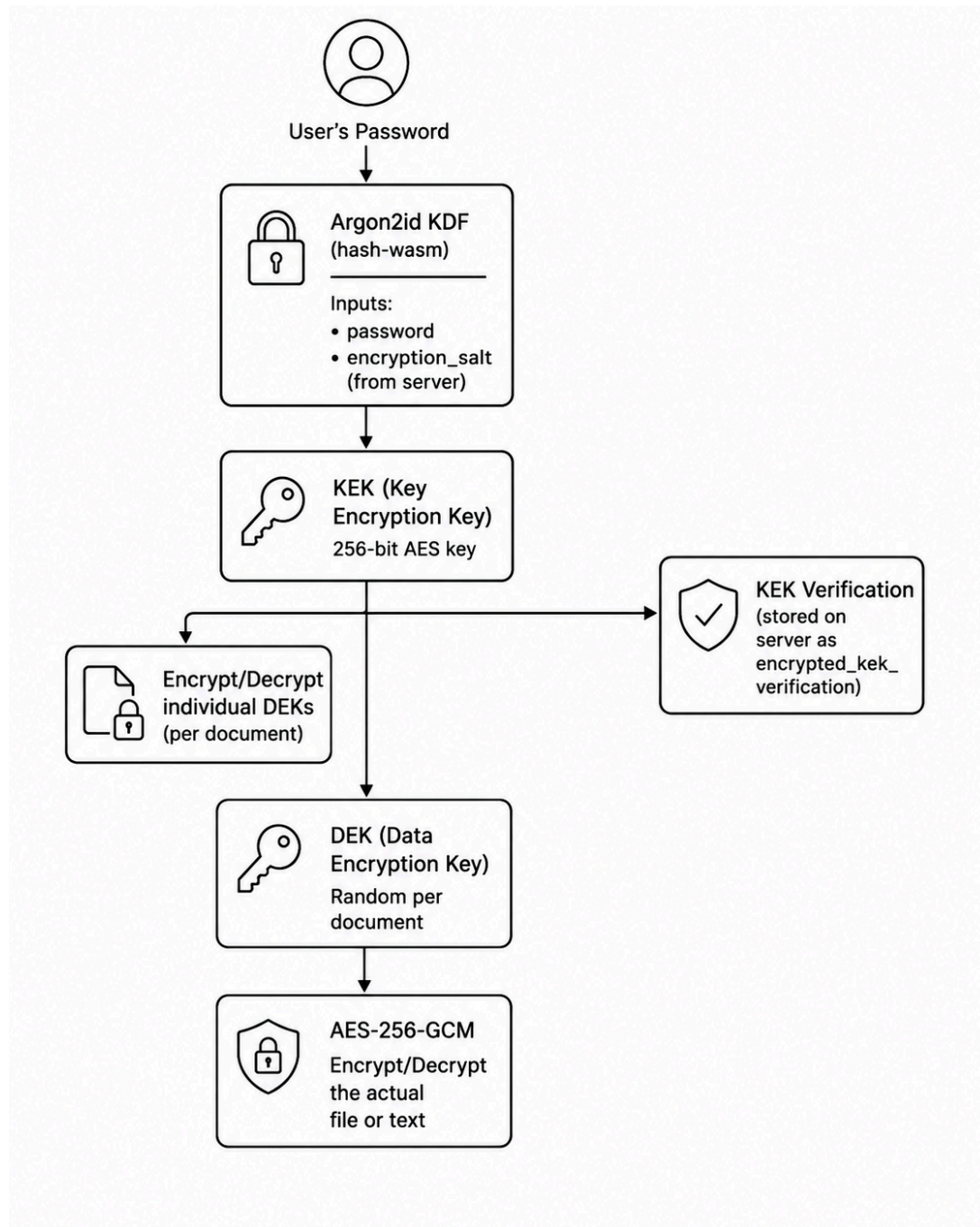
Data Flow Summary:

1. User enters data in the Angular frontend.
2. Sensitive fields are encrypted client-side
3. Encrypted payloads are sent to the Spring Boot backend via REST API with JWT authentication.
4. Backend stores encrypted blobs
5. Files are stored in MinIO; metadata goes to PostgreSQL.
6. On retrieval, encrypted data is sent back to the client and decrypted in-browser.

Tech Stack

Layer	Technology
Frontend	Angular
Client Crypto	Web Crypto API (AES-256-GCM) + hash-wasm (Argon2id)
Backend	JAVA, Spring Boot
Database	PostgreSQL
Migrations	Flyway
Object Storage	MinIO (S3-compatible)
Auth Tokens	JWT (io.jsonwebtoken)
Email	Spring Boot Starter Mail
Build Tool	Maven (mvnw wrapper)

Security and Encryption Model



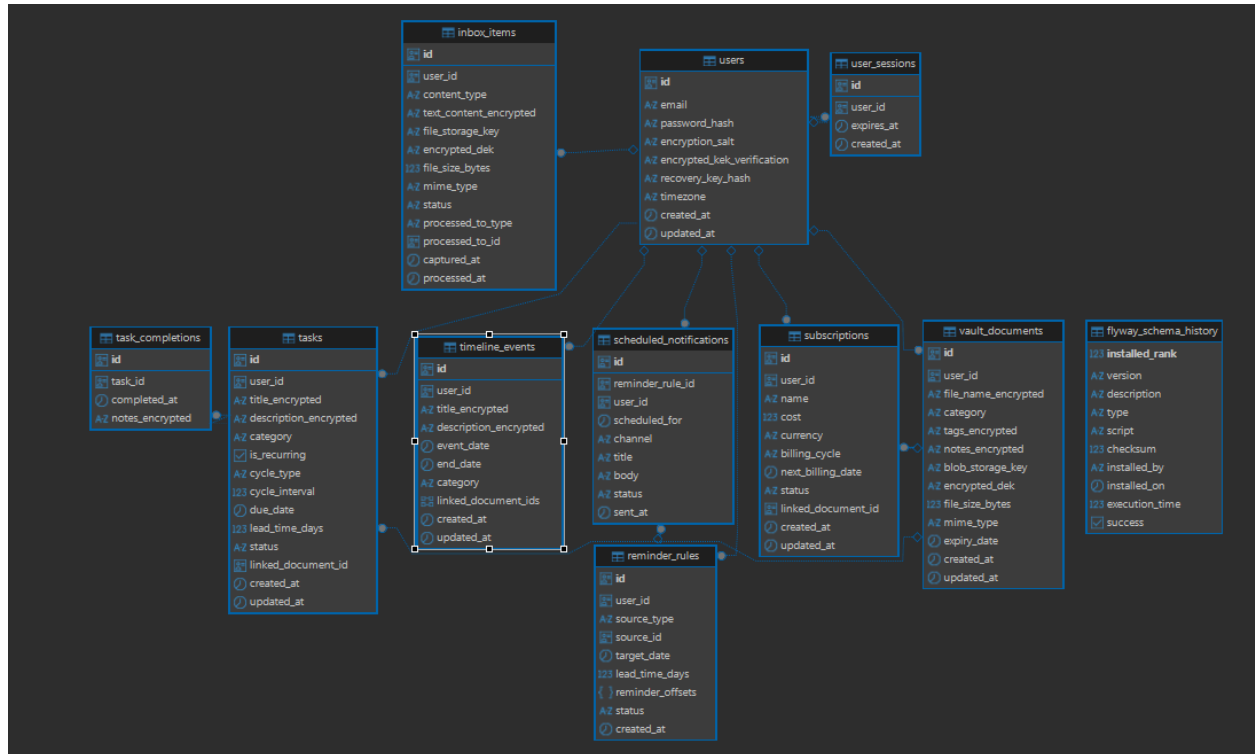
End-to-End Encryption Flow

1. User enters their password in the application.
2. The client retrieves the user's unique encryption_salt from the server.
3. The client derives a 256-bit Key Encryption Key (KEK) using Argon2id (hash-wasm) with:
 - a. Password
 - b. encryption_salt
4. The client generates a random 256-bit Data Encryption Key (DEK) for the document being uploaded.
5. The client generates a random Initialization Vector (IV/Nonce) for AES-256-GCM.
6. The client encrypts the file (or text) using AES-256-GCM with:
 - a. DEK as the encryption key
 - b. IV as the nonce
7. The client encrypts the DEK using the KEK, ensuring the DEK is never stored in plaintext.
8. The client uploads the encrypted data to the server, including:
 - a. Encrypted file (ciphertext)
 - b. Encrypted DEK
 - c. IV (Nonce)
 - d. Encrypted metadata
9. The server stores only encrypted data and never has access to the user's password, KEK, DEK, or plaintext files.
10. When the user wants to access a document, they enter their password again.
11. The client retrieves the encryption_salt and derives the same KEK using Argon2id.
12. The client verifies the KEK by decrypting the stored encrypted_kek_verification value. If verification succeeds, the password is correct.
 - a. The client downloads the following from the server:
 - b. Encrypted file
 - c. Encrypted DEK
 - d. IV (Nonce)
13. The client decrypts the DEK using the KEK.
14. The client decrypts the file using AES-256-GCM with the decrypted DEK and IV.
15. The original plaintext file is recovered, and all encryption and decryption operations occur locally on the client device.

What gets encrypted Client-Side

Data	Encrypted Field
Vault document files	File bytes encrypted with per-document DEK
Vault file names	file_name_encrypted
Vault notes	notes_encrypted
Inbox text content	text_content_encrypted
Inbox file attachments	File bytes encrypted with per-item DEK
Task titles	title_encrypted
Task descriptions	description_encrypted
Task completion notes	notes_encrypted
Timeline event titles	title_encrypted
Timeline event descriptions	description_encrypted
Subscription names	name (stored in plaintext on server — not encrypted client-side)

Database Schema



User Manual

Getting Started

Sign up and Login

1. Go to <http://localhost:4200/register>
2. Enter your email and password. Since this is a privacy-focused app, it only stores your email address and password. No additional personal information is collected or stored.
3. After that you will get a recovery key, it is the only way to recover your vault if you forget your password.
4. App will redirect to login page, enter credentials to go to dashboard.

Dashboard

This is the entry page of the app. Here you can see stats of:

1. Pending Tasks
2. Active Subscriptions total monthly
3. Unprocessed items in staging area
4. Quick capture panel to drop your thoughts.

Quick Capture

Capture now, organize later.

How to use:

- Text: Type a note, click "Capture"
- File: Drag and drop (not working) or click to upload a file

Organize:

- Send to vault
- Send to task
- Send to subscription
- Delete

Vault

Store important documents securely. All files are encrypted before upload.

Upload a document:

1. Click "Upload Document"
2. Select a file
3. Choose a category.
4. Add tags and notes (optional)
5. Set an expiry date if needed (auto-generates reminders)
6. Upload

Subscription

Add a subscription:

1. Click "Add Subscription"
2. Fill in:
 - Name
 - Cost and currency
 - Billing cycle (Monthly, Quarterly, Semi-annual, Annual)
 - Next billing date
 - Status (Active, Paused, Cancelled)
3. Save

Manage:

- View all subscriptions with upcoming renewal dates
- Edit details anytime
- Delete subscriptions you no longer want
- Link receipts/documents from your Vault

Tasks

There are two type of task you can add:

1. Not Recurring task (one off task)
2. Recurring task

Add a task:

1. Click "Add Task"
2. Fill in:
 - Title and description
 - Category
 - Due date
 - Lead time (days before due date to get reminders)
 - Toggle "Recurring" if it repeats
 - Set cycle: Days, Weeks, Months, Years
 - Set interval (e.g., every 3 months)

3. Save

Timeline

Keep a private record of your life events.

Add an event:

1. Click "Add Timeline Event"
2. Fill in:
 - Title and description
 - Event date
 - End date
 - Category (Career, Education, Health, Legal, Personal, Financial, Property)
 - Link relevant Vault documents
3. Save